



Seeking Pleasure, Finding Trouble: Functions and Dysfunctions of Trait Sensation Seeking

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Abstract

Purpose of Review Sensation seeking, a trait-like preference for intense experiences with a disregard for their potentially aversive consequences, is a risk factor for the development of substance use disorders and other hazardous activities. However, there remain several open questions regarding its neurophysiological characteristics, its precise effect on decision-making, and the clinical management of high sensation-seeking individuals.

Recent Findings We review evidence, including recent neuropharmacological and neuroimaging work, which demonstrates a neurophysiological basis for the construct, particularly regarding reward-related and punishment-related processes. We then evaluate evidence for a relationship between sensation seeking and behavioral measures of risk taking, which supports a consistently modest increase in risk preference with higher trait sensation seeking. These small effect sizes imply the need both for more sensitive behavioral measures, as well as novel theoretical insights into when and why high sensation-seeking individuals might show a preference for risky options. Predictions derived from different theoretical approaches, including subjective utility, reinforcement learning, active inference, and contextually dependent risk taking, are outlined.

Summary Novel theoretical approaches might help to derive more specific measures of sensation seeking via the generation of new paradigms. They might also clarify adaptive and harmful aspects of the decision-making style of high sensation seekers, which might help to develop and refine new clinical interventions.

Keywords Sensation seeking · Risk taking · Foraging · Active inference

Introduction

Drugs and other hazardous recreational activities (e.g., gambling) are widely available in many countries throughout the world. Despite their prevalence, considerable evidence exists to suggest that only a subset of individuals are particularly susceptible to developing substance use disorders (SUDs: [1]). Of the risk factors that might predispose individuals to dysregulated or compulsive use of substances or gambling, there has been considerable interest in personality traits. In particular, impulsiveness—the predisposition for behavior which is risky, poorly planned, and/or favors immediate small rewards over larger, delayed rewards [2]—has received

considerable empirical support as a potential risk factor [3, 4]. From a behavioral economic perspective, drug use is conceptually related to impulsivity, given that drugs can provide not only an immediate, intensely rewarding experience but also risks of harm [5]. Hence, understanding trait-like variation in what determines decisions to seek such experiences is central to understanding and predicting the course of addiction in at-risk individuals.

The concept of impulsivity is multifaceted. It can be measured by several self-report scales [6, 7], each providing measures of different subcomponents, as well as a variety of behavioral measures [8–10]. Within the factor structure of impulsivity self-report scales, dimensions reflecting reward sensitivity can emerge: a trait-level factor reflecting approach motivation and subjective hedonic responses to rewarding events [11, 12], which appears distinct from disinhibition-related aspects of impulsivity [13]. Note that enhanced sensitivity to reward does not imply that all decisions to obtain large, intense rewards should be impulsive [14]—although, in practice, associations between construct

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measures of reward sensitivity and impulsivity can be observed [12, 15]. While reward sensitivity is quite a general concept, relevant for a number of psychiatric disorders including schizophrenia [16] and depression [17], the related construct of sensation seeking (SS) represents the coupling of a preference for “varied, novel, complex, and intense sensations and experiences,” with a potential disregard for the aversive—“physical, social, legal, financial”—consequences which might result [18, 19•]. It is commonly measured via self-report using the Zuckerman Sensation Seeking Scale (SSS: [20]), the Sensation Seeking subscale of the UPPS-P [7], or the Arnett Inventory of Sensation Seeking (AISS: [21]). The SSS includes four subscales: thrill and adventure seeking, experience seeking, disinhibition, and boredom susceptibility [20]. To our knowledge, the hierarchical relationship between SS and other facets of reward sensitivity has not been widely evaluated (but see [22]).

In the present review, we start by describing data which serve to validate the SS construct, including self-report, behavioral, and neural data. We then describe empirical evidence supporting a relationship between risky choice and SS, in the context of different theoretical approaches to understanding decision-making. We then evaluate theories of risk taking in more detail, and in the final section, we explore how predictions derived from these theories might be explored in future work.

Validation of Sensation Seeking

SS has received considerable validation as a construct distinct from other aspects of impulsivity and reward sensitivity. Scales measuring SS can have reasonable psychometric properties [20, 23]. SS can be distinguished from other forms of impulsive behavior including urgency or lack of premeditation [7] and from novelty seeking [19•]. Further work has demonstrated increased self-reported SS across a variety of groups relative to control individuals, including individuals with SUDs [24•], pathological gamblers [25], downhill skiers [26, 27], skydivers [28], and chess players [29]. In addition, higher SS predicted lower abstinence within the trial of an intervention for heavy drinking smokers [30]. In one particularly significant study, Ersche and colleagues [24•] demonstrated increased SS in individuals with SUDs (stimulant users), but not their first-degree relatives, relative to control participants. By contrast, self-reported impulsivity, as measured by BIS-11, was elevated in both the SUD group and their first-degree relatives compared to healthy controls. We note two intriguing implications of this study: First, the interaction of different facets of impulsivity may be most predictive of the development of SUDs, as opposed to any particular facet; second, SS may be susceptible to influence by an individual’s experience (e.g., early

life experiences, exposure to drugs). However, while some genetic predisposition to impulsive traits including SS has been established [31], the distinct role of genetics versus experience in the development of complex traits such as SS remains to be determined.

Further work has sought to identify neurobiological substrates of SS. While relationships between resting state functional connectivity and SS have been identified [32], a larger body of work has examined neural responses coupled to rewards or punishments in order to understand what neural processes might drive risky choices in high SS individuals. Several findings, examining feedback-related event-related potentials (ERPs), are consistent with a reward or punishment *hypo-sensitivity* model, in which high SS is associated with diminished neural responsivity to or discrimination of feedback, including rewards and losses [33, 34]. Within this model, high SS individuals would seek larger rewards to obtain the same level of stimulation as small rewards would provide to low SS individuals, and would disregard risks. Further evidence has been obtained using functional magnetic resonance imaging (fMRI): For example, skydivers with a low anticipatory anxiety response also showed a blunted neural response to aversive cues in the amygdala [35]. Nevertheless, there is also evidence for a *hyper-sensitivity* model, in which high SS individuals would show enhanced responses to reward outcomes or expectancy [36, 37•]. A general theme across this literature is a focus on dopamine as a biological substrate of reinforcement, given the relationship between dopamine, motivation, and reward [38, 39]. Dopamine receptor density and/or release might then differ across levels of SS, mediating the effects of reinforcement on behavior [19•].

Other evidence suggests that the experience of reward may be qualitatively different in high SS individuals, and, in particular, might be tuned to different types of reinforcer, as opposed to a simple hyper/hypo-sensitivity. One striking example has been subjective responses to a stimulant (d-amphetamine: [40]): a clear positive affective response in high SS individuals, versus a negligible affective response in low SS individuals. Further work has examined the intensity of stimulation, observing marked individual variation in responses to a shock: High SS individuals were willing to forgo monetary rewards to obtain small electric shocks [41]. In this context, it is notable that a prominent model of demand [42] also includes an intensity parameter—reflecting the fact that there can be variation in preference for the magnitude of a reinforcer—which is potentially independent of the cost an individual might be willing to incur in order to obtain such a reinforcer. Together, these findings argue against a simple unitary model of reward sensitivity: One might be very motivated to obtain a reward, but not necessarily consume a great deal of it; and one might forgo one type of reward for another across a dimension of reinforcer intensity.

SS is defined in terms of a preference for intense rewards [18], and these studies suggest how intensity might be operationalized experimentally.

Finally, rodent models relevant for SUD research demonstrate marked variation in hedonic responses and reward seeking, including variation related to locomotor activity within a novel environment [43•]. Such variation may represent a translational substrate of SS, and provide potential avenues for cross-species validation of phenotypic characteristics of SS and to explore the neurobiological substrates of risk factors for SUDs.

Together, these findings establish the existence of SS as a valid axis of individual differences, with a capacity to reflect and predict risky behavior including drug use. SS is typically measured using various self-report scales, but more recent work has demonstrated a potential neurophysiological basis for the construct.

Sensation Seeking as a Predictor of Risk Preference

Understanding how high SS individuals come to make the choices they do could provide insight into who might be at risk of dysregulated substance use, and why. Consequently, there have been numerous evaluations of the relationship of SS with risk preference. Broadly speaking, there are two primary theoretical frameworks for thinking about these relationships. The first is to evaluate economic decision-making, and preferences for particular options, in terms of subjective utility. The utility of an option would be constructed via weighting of the expected value and likelihood of positive and negative outcomes by an individual's subjective preferences and beliefs. Seen within this framework, SS might be seen as a stronger preference for rewarding outcomes relative to an aversion to negative outcomes (i.e., a reduction of loss aversion). Thus, a sensation-seeking individual might be more susceptible to information about potential upsides of a decision over information about potential losses. Whether or not a decision-making policy is adaptive (e.g., yields the highest net reward) can depend on the task and is potentially independent of risk preference, with some paradigms penalizing [44] and others revealing advantages of risk taking [4, 46].

Alternatively, the utility of decision options can be weighted by a non-linear function of expected value. For example, a monetary reward of \$100 might be weighted as more than twice as valuable as a \$50 reward. Such an inference can be made on the basis of the fitting of decision-making models to participants' pattern of choices—with a key assumption being the transitivity of preferences [47]. Human decision-makers are often thought to show concave utility functions for rewards [48, 49], with smaller (although not tiny [50]) rewards showing a

relatively greater utility than larger rewards. This pattern might be inverted in high SS individuals, who would over-value large relative to small rewards. Finally, individuals may weight probabilities differently: For example, high SS individuals might over-weight the likelihood of an uncertain reward occurring.

An economic framework of this sort might effectively account for reported relationships between SS and prior demonstrations of risky behavior, with greater SS be associated with greater risky choice [51–54]. In addition, risk preferences may be modulated by dopamine: Norbury and colleagues observed a modulation of risk taking by cabergoline, a D2/D3-receptor agonist, in a manner which depended on trait SS [55].

The second broad approach is to consider what representations of potential rewards or punishments in the environment might drive behavior. A very simple starting point would be the idea that learning from rewards would be enhanced in high SS individuals, but learning from punishments would be diminished. A key difference between learning and economic approaches concerns the nature of their underlying assumptions. In economic paradigms, participants are given access to the nature of potential outcomes and their probabilities, and choices are generally assumed to be made solely on the basis of the information available (albeit perhaps degraded), as well as the participant's current motivational state. By contrast, a learning-based account would emphasize the role of prior experiences in determining preferences, as well as different representations of predictive cues, actions, and outcomes. This opens a variety of avenues for other mechanisms to affect choice which are not straightforwardly available to the economic account, and these may be responsible for heightened risk preference in high SS individuals.

For example, the presence of cues which are predictive of rewarding or punishing outcomes might bias choices in ways which are not consistent with their utility [56]. Moreover, there is evidence of context-specific influences on choice, such that risk preference might be influenced by local contextual factors (beyond expected utility) such as the type of feedback [57, 58]. Furthermore, the state of the individual during encoding may influence the expected utility of the outcome: Experiencing an outcome during a deprived state can lead it to acquire greater value [59]. The relevance for SS is that intense experiences that high SS individuals can find appealing are generally not those experienced within a laboratory. To address this experimental limitation, relevant stimuli (e.g., drug cues [60]) or contexts (e.g., casinos/bars [22, 61, 62]) can be used within studies of substance use to provide more motivationally relevant stimuli.

Across groups of distressed and healthy young adults, we observed a relationship between preference for risky versus safe options and the experience seeking scale of

the Zuckerman SSS scale [37•]. In this study, the risky options had been previously associated with uncertain wins and losses during a card-guessing task, so participants are expected to have learned their value from experience. By contrast, the safe options were provided explicitly. Recently, Garcia and colleagues have more systematically examined preference between options whose outcome probabilities were learned by experience versus where outcome probabilities were explicitly presented [63•], but where the two option classes were otherwise equivalent. This study may have implications for the interpretation of our findings [37•]. First, Garcia and colleagues showed that experientially learned options were generally neglected relative to explicit options: Our observation that high SS individuals showed greater preference for the risky options learned by experience [37•], relative to low SS individuals, suggests that a neglect of experience may be moderated by individual differences including SS. While this proposal remains to be confirmed within an appropriately controlled design, if it is correct, it may in turn have implication for risky behaviors: Health or risk-related information which is presented explicitly (e.g. health warnings [64]) may be relatively ineffective in high SS individuals. Similar, whether the distinction between risky (i.e., uncertain options where outcome probabilities are explicitly presented or where the participant can estimate them precisely) and ambiguous choices (i.e., uncertain options where the outcome probabilities are unknown [65]) is relevant for sensation seeking has not been widely investigated to our knowledge. One study demonstrated an effect of SS on a “Game of Dice” task in which outcome probabilities are explicitly presented, but not on the Iowa Gambling Task, where they are learned during the task [66].

Regardless of one’s preferred explanatory system, empirical effect sizes which map behavioral measures, such as laboratory-based choice tasks, onto self-report measures of SS, impulsivity, or similar constructs are generally consistent but small [51–53, 67•, 68]. This has several implications for the development of theoretical accounts of SS. While we might be confident of the validity of SS as a construct, behavioral measures would allow us to move more decisively beyond participants’ self-reports, and generate a wide-ranging model of the preferences of high SS individuals revealed by *choice*. Modest effect sizes imply other factors beyond SS play a role in determining risk preference, and also that a domain-general preference for larger but riskier rewards does not provide a particularly effective account of SS. This empirical disconnection might be resolved in two ways. First, we could try to identify risk-based measurements with the strongest effect sizes and build on these to develop more sensitive measures. An important recent innovation here is the use of a Bayesian adaptive design optimization [69], which may provide more precise and rapid assessment of individual differences including utility weighting

or probability discounting. Second, we could attempt to generate novel paradigms based on insights from different theoretical models. We sketch such theoretical approaches in the following section.

Foraging and the Value of Information

Describing SS in terms of risk preferences is consistent with the definition of the SS construct, and is directly supported by empirical behavioral and self-report data. However, describing SS in this way does not explain *how* values are constructed, nor *why* individuals might show different appetite for high intensity rewards (i.e., what is the beneficial “competence” [70] provided by risk taking). In terms of the former point, the neuroimaging or psychopharmacological work in this area which has already been conducted, as reviewed earlier in the text, might represent avenues toward developing a mechanistic, neurobiological grounding for the construct.

The second point is likely to be more contentious. For the purposes of this review, we will address two different types of explanation. The first explanation is rooted in ethology, and aims to account for risk preferences across different species. For example, it has been hypothesized that animal risk preferences may reflect different foraging strategies in response to deprivation [71], or different resource-dependent mating strategies [72]. Such accounts would provide an explanation for why SS should vary so much across individuals: Whether or not a given set of risk preferences is optimal may depend on a variety of contextual factors, which may favor certain individuals at certain times or places, and others at other times. Similarly, early environmental stress might influence an individual’s impulsiveness [73] or risk preferences, perhaps to prepare them for the environments they might expect to face. Notably, Bickel and colleagues [74] demonstrated that heroin users generally preferred to purchase heroin from a variable supplier rather than a more consistent supplier if asked to imagine heroin deprivation rather than satiety. This finding is consistent with the notion that risk preference can be influenced by deprivation state, as well as demonstrating relevance for drug seeking behavior.

A second type of explanation is rooted in the “active inference” framework [75], which has emphasized the role of the information gathering and exploration, in addition to optimizing action policies based on expected utility. All learning models, including reinforcement learning (RL), require consideration of the balance between exploration and exploitation—namely, exploiting the location of known sources of rewards, versus exploration of new, previously untapped sources [76–78]. As with risk taking, the optimal balance of exploration and exploitation depends on the

environment [79], and experience within an environment may help to determine how best to set this balance.

A key contribution of the active inference framework is to emphasize information as a currency, in addition to reward. Here, agents are preoccupied with minimizing expected free energy by building suitably predictive models of the environment [75]. One important point about high intensity rewards is that they are generally unusual, and that they often require a degree of exploration and/or effort to obtain. Notably, if they were not rare (i.e., high intensity rewards were straightforwardly available), it seems likely that the reward system would adapt [80–82], and what was previously considered “high” would be more coded as more moderate. Thus, the relative, rather than absolute, coding of rewards, implies that rewards with a high subjective utility would be necessarily infrequent and out-of-the-ordinary. To obtain such rewards would then require exploration and also potentially risk taking.

Briefly setting aside aspects of SS relating to the sensitivity to reward or preferences for different types of reward, both risk taking and exploration are examples of behavior which is likely to result in information transfer from the environment to the individual. Although agents aim to minimize expected free energy in an effort to obtain reliable, preferred rewards, information seeking is necessary in order to build an internal model of an environment which is sufficiently predictive [75]. Thus, preference for informative options or environments [83], independent of utility, might drive risk preference and/or directed exploration at least while an agent is uncertain about aspects of an environment, which might help to provide insight into its deeper causal structure. For example, while a clearly discriminable cue might be an uncertain predictor of a salient outcome (e.g., whether a fruit is ripe or not), there may be less discriminable cues (e.g., its exact hue or shape) which might be more effective predictors, but only be apparent following further examination. Accordingly, eyetracking can reveal that orienting toward cues which are unreliable predictors of outcomes can be increased relative to better predictors [84]. Similarly, sign-tracking behavior, thought to be performed in order to obtain information about the causal structure of the environment [85], can be enhanced to uncertain reward-predictive cues in rodents [86]. Along these lines, we observed individuals with greater self-reported SS showed a widespread pattern of increased activation coupled to uncertain reward-related cues using fMRI [37•], particularly in regions relevant for attention. Although we should be mindful of reverse inference [87] when claiming these data might support an altered attentional process in SS, the data nevertheless suggest that fMRI might be used, in addition to eyetracking, to test the effect of SS on uncertainty-related attention and learning.

Together, this framework allows consideration of the notion that SS may be related to information seeking or

curiosity, and that this may exist independently of reward sensitivity. In practice, high trait SS might represent the culmination of high levels of information seeking with elevated reward sensitivity. It is possible to imagine information seeking without reward sensitivity, for example in the case of seeking information about sources of threat [88]. So while a preference for intense rewards is central to the definition of SS, information seeking may be a further dimension which is needed to distinguish it from other forms of trait variation in reward-related behavior.

Predictions and Future Directions

So far, we have examined three theoretical approaches to understanding SS. First, SS might be described in terms of a relative insensitivity to small rewards, and to punishments, relative to large, intense rewards. Because this model can provide straightforward explanations for observed patterns of choice and neural data, we would see this as a kind of benchmark approach against which other theoretical approaches could be tested and developed. An important aspect of this approach is whether reward or punishment is coded in an absolute or relative scale [80, 82, 89]: specifically, the extent to which reward coding is recalibrated in response to environmental changes in the range of rewards available. To our knowledge, it is not well understood whether this recalibration might differ across the range of SS or across different types of reinforcer. Further areas of investigation might concern how option utilities are constructed across a range of SS, and so what might specifically influence the preferences of high SS individuals. Such investigations might also consider the difference between risk and ambiguity [65], and consider experiential learning of expectancies [63•], perhaps within environments which might be more relevant for intense experiences (e.g., [61]).

A second model would emerge from the idea that risk taking might be adaptive for a particular individual in response to environmental changes. For example, deprivation might require an individual to forage in more risky environments [71] (but see [90]). In this light, high SS might be an extreme and/or contextually inappropriate manifestation of an otherwise adaptive response to an environmental challenge. For example, an experience of deprivation might induce a pattern of risk-taking behavior which does not end once deprivation ends, or perhaps interacts in an extreme way with non-natural reinforcers such as drugs. Key explanatory possibilities afforded by this account include domain-specific risk taking (e.g., risk taking for one type of reinforcer and not another) and/or context-specific risk taking (e.g., risk taking prompted by a particular motivational state). To our knowledge, these remain to be systematically evaluated in the context of SS.

A final model entertains the idea of information as a currency, independent of reward. Within this account, SS might be distinguished from differences in reward sensitivity by an enhanced curiosity about environmental structure, which would promote seeking intense rewards in unusual or novel environments. From this point of view, SS would exist along a natural continuum of exploration/exploitation—essentially being an unusual but potentially adaptive foraging strategy. Key testable predictions of this account relate to preferences for directed versus random exploration [91–93] and for orienting to cues which predict outcomes unreliably [84]. It is likely that a preference for intense rewards would also be needed to model the full SS phenotype, but this model might also provide more focused predictions regarding information seeking. Intriguingly, there is evidence that experience of boredom increases exploration [94], so we might predict that the boredom susceptibility component of SS may be related to exploration.

Thus far, we have considered SS in terms of an *individual's* preference for reward or information. A further important aspect includes the social dimension, and in particular, the extent to which risk taking can be contingent on the social milieu [95], and also the extent to which foraging might proceed socially [96, 97]. In terms of the latter, given that information about the location and nature of rewards is crucial to successful foraging, the extent to which such information can be shared across a social group might be important for understanding the function of SS. To our knowledge, this has not received detailed examination.

One crucial theoretical point is the extent to which SS is goal-directed. Given that it is assessed using self-reports, individuals with high SS describe an awareness of their preferences and motivations, and in particular that they perceive that they intentionally seek the rewards and environments they do. Note that a specific preference for high-information and/or novel environments contrasts with the establishment of repetitive, habitual behaviors [98], and their role in SUDs and behavioral addictions [99]. One resolution for this idea is that SS might act as a risk factor for the *onset* of maladaptive behaviors, but that other risk factors might predispose toward *repetitive* maladaptive behaviors [99]. While there is evidence of reduced goal-directed control in high SS individuals [100], such an association remains to be evaluated more widely.

Notably, while we have generally sketched these models independently, they should not necessarily be taken to be in opposition. Rather, they are distinct and potentially complementary approaches to explanation. For example, insensitivity to small rewards is an algorithmic explanation which could be applied to a particular motivational system to serve a given computational purpose. Nevertheless, the different models imply different predictions which could be tested using distinct methodologies.

Conclusion

Sensation seeking (SS) is a disposition toward seeking risky, intense sensations, and can be associated with maladaptive behaviors such as substance use or gambling. In the present work, we have described empirical evidence validating SS as a distinct construct, and its possible neurobiological basis. We have also pursued different theoretical accounts of what function seeking risky rewards might serve, beyond algorithmic models based in reinforcement learning or economic decision-making: namely, ethological accounts emphasizing particular benefits of risk seeking in certain contexts (e.g., deprivation); and recent active inference accounts emphasize the pursuit of information and consequent preferences for uncertain environments. Experimental strategies to delineate different accounts are suggested. Overall, a broader perspective regarding the benefits and harms of trait SS may help to provide a framework for considering interventions to avoid the maladaptive, but emphasize the adaptive, consequences of high trait SS.

Glossary

<i>Active inference</i>	An approach to computational learning which emphasizes the construction of probabilistic models of an environment which minimize free energy i.e., how surprising and informative events are.
<i>Adaptive coding /range adaption</i>	The capacity of neural systems to adjust to the range of stimulation experienced within a given context.
<i>Goal-directed/ habit</i>	Goal-directed behavior is sensitive to its expected consequences: Thus, an individual should immediately reduce responding for non-preferred options. By contrast, habitual behavior is not sensitive to its consequences, but automatically elicited by an environment or stimulus following training.
<i>Exploration/ exploitation</i>	The balance between seeking new sources of reward (exploration) versus seeking to consume reward from known sources (exploitation). This is a crucial consideration in the development of RL models, and for understanding foraging behavior.
<i>Reinforcement learning (RL)</i>	RL models are algorithmic models of learning which represent relationships of stimuli and/or actions with rewards or punishments. Models generally include a prediction error learning mechanism, in which differences between expected and experienced reinforcement are used to update expectations.

Utility

The subjective value of a particular option. Individuals differ in their preferences for different commodities or experiences, and their choices are expected to follow these subjective preferences. Utility can be modeled algorithmically by weighting different options via a non-linear function

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Declarations

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